22-10-2024

RANDOM VARIABLE

- Any fn that assign a numerical value to each the possible outcome
- Denoted by capital letters.
- Two type: 1. Discrete - take only discrete values.
  2. Continuous - take value in an intervals.
- If the numerical values assumed by a variable are the result of some chance factors, so that a particular value cannot be exactly predicted in advance, the variable is then called a random variable. A random variable is also called 'chance variable' or 'stochastic variable'.
- A r.v. is any function that assigns a numerical value to each possible outcome.
- R.V are denoted by capital letters, usually, from the last part of the alphabet, for instance, X, Y, Z, etc.

CONTINUOUS AND DISCRETE RANDOM VARIABLES

- A continuous R.V is one which can assume any value within an interval, i.e., all values of a continuous scale.
- For eg: (i) the weight (in kg) of a group of individuals
  (ii) heights of a group of individuals.
- A discrete R.V is one which can assume only isolated values.

For eg: (i) the number of heads in 4 tosses of a coin is a discrete r.v. as it cannot assume values other than 0, 1, 2, 3, 4.

(ii) the no. of aces in a draw of 2 cards from a well shuffled deck is a r.v.s as it can take the values 0, 1, 2, only.

DISCRETE PROBABILITY DISTRIBUTION FUNCTION (PROBABILITY MASS FN)

Let a random variable X assumes values $x_1, x_2, \ldots, x_n$ with probabilities $P_1, P_2, \ldots, P_n$ respectively where PDF

$P(X = x_i) = P_i \geq 0$ for each x_i and

$P_1 + P_2 + \ldots + P_n = \sum P_i = 1$.

Then,

X :	x_1	x_2	.....	x_n
P(x):	P_1	P_2	.....	P_n

is called the discrete probability distribution of X.

Gives the info about how probability is distributed.

CONTINUOUS PROBABILITY DISTRIBUTION FUNCTION (CONTINUOUS PROBABILITY DENSITY FUNCTION) PDF

If a random variable is continuous its probability distribution is called continuous probability distribution function.

X is a continuous random variable, then the PDF (Probability density function), f(x) satisfies the following properties:

(i) $f(x) \geq 0$ (f(x) may vary).

(ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

PDF can be used to determine the probabilities of continuous random variable.

(i) $P(x \geq 0) = \int_{0}^{\infty} f(x)\,dx$

(ii) $P(x \leq 0) = \int_{-\infty}^{0} f(x)\,dx$

(iii) $P(a \leq x \leq b) = \int_{a}^{b} f(x)\,dx$

Definition:

DISTRIBUTION FUNCTION

- If X is a ranom variable ~~then~~ then $P(X \leq x)$ is called distribution function and is denoted by $F(x)$.

$$F(x) = P(X \leq x) = \begin{cases} \sum_{x=0}^{x} P(x), & \text{if } X \text{ is discrete random variable (DRV)} \\ \int_{0}^{\infty} f(x)\,dx, & \text{if } X \text{ is continuous RV (CRV)} \end{cases}$$

PROPERTIES

1. $F(x) \geq 0$
2. $F(\infty) = 1$
3. $F(x)$ is non-decreasing function. (either ↑ or constant).

EXPECTATION OF X (MEAN OF RANDOM VARIABLE)

- Let X be a discrete random variable with PDF (or PMF) $P(x)$. Then the mean or expectation of random variable X is given by

$$E(X) = \sum_{x=0}^{n} x\,P(x)$$

- If X is a continuous random variable with PDF $f(x)$, then,

$$E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$$

PROPERTIES

1. $E(k) = k$, where k is a constant.
2. $E(kX) = k\,E(X)$
3. $E(aX+b) = a\,E(X) + b$, where a and b are constant.
4. $E(x^2) = \Sigma x^2 P(x)$, if X is a discrete random variable
   $= \int x^2 f(x)\,dx$, if X is a continuous random variable.

VARIANCE OF X

$E(X-\bar{X})^2$ is called the variance of X, where $\bar{X}$ is mean or expectation of X. ~~also~~

$$Var(X) = E(x^2) - [E(x)]^2$$

PROOF

$$Var(X) = E(X-\bar{X})^2$$
$$= E(X^2 - 2X\bar{X} + \bar{X}^2)$$
$$= E(X^2) - 2\bar{X}E(X) + E(\bar{X}^2)$$
$$= E(X^2) - 2(E(X))^2 + (\bar{X})^2 \qquad (\because \bar{X} = E(X))$$
$$= E(X^2) - 2(E(X))^2 + (E(X))^2$$
$$Var(X) = \underline{E(X^2) - [E(X)]^2}$$

PROPERTIES

1. $Var(k) = 0$, where k is a constant.
2. $Var(kX) = k^2 Var(X)$
3. $Var(aX+b) = a^2 Var(X)$

STANDARD DEVIATION

It is the positive square root of variance.

$$SD = \sqrt{Var(X)}$$

Q: A random variable X has the following PMF.

Value of X:	-2	-1	0	1	2	3
P(X) :	0.1	k	0.2	2k	0.3	k

Find the value of k.

ans:

$$\Sigma P_i = 1$$
$$0.1 + k + 0.2 + 2k + 0.3 + k = 1$$
$$0.6 + 4k = 1$$
$$k = \frac{0.4}{4} = 0.1 //$$

Q: Given that $f(x) = kx^4e^{-x}$, $0<x<\infty$ is a PDF. Then determine k.

Ans:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{\infty} kx^4e^{-x}\,dx = 1$$

$$k\int_{-\infty}^{\infty} x^4e^{-x}\,dx = 1$$

$0<x<\infty, \infty$

$$\Rightarrow k\int_{0}^{\infty} x^4e^{-x}dx = 1 \qquad \left(\int_0^\infty e^{-x}x^{n-1}dx = \lceil n\right)$$

$$\lceil n = (n-1)!$$

$$k\lceil\rceil_0^\infty$$

$$k\int_0^\infty e^{-x}x^{5-1}dx = 1$$

$$k\lceil 5 = 1$$

$$k(5-1)! = 1$$

$$\Rightarrow k\,4! = 1$$

$$k = \frac{1}{4!} = 0.04167$$

Q: Find the mean and variance of the random variable with PDF $f(x) = k.(1-x^2)$ $0<x<1$.

ans:

To find k,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\Rightarrow \int_0^1 k(1-x^2)^1dx = 1$$

$$\Rightarrow k\left[\frac{(1-x)^3}{3}\times -1\right]_0^1 = 1$$

$$\Rightarrow k\left[x - \frac{x^3}{3}\right]_0^1 = 1$$

$$\Rightarrow k\left[1 - \frac{1}{3}\right] = 1$$

$$\Rightarrow k = \frac{3}{2}$$

Mean:

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$E(X) = \int_{0}^{1} x \times \frac{3}{2}(1-x^2)\,dx$$

$$= \frac{3}{2}\left[\int_0^1 x - x^3\,dx\right]$$

$$= \frac{3}{2}\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1$$

$$E(X) = \frac{3}{2}\left[\frac{1}{2} - \frac{1}{4}\right] = \frac{3}{2} \times \frac{1}{4} = \frac{3}{8}$$

Variance $= E(x^2) - (E(x))^2$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

$$= \int_0^1 x^2 \times \frac{3}{2}(1-x^2)\,dx = \frac{3}{2}\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1$$

$$E(x^2) = \frac{3}{2}\left[\frac{1}{3} - \frac{1}{5}\right] = \frac{3}{2} \times \frac{2}{15} = \frac{1}{5}$$

$$Var(X) = \frac{1}{5} - \left(\frac{3}{8}\right)^2 = 0.059375$$

$$Var(X) = \frac{1}{5} - \frac{9}{64} = \frac{19}{320}$$

Q: Obtain the distribution function and mean of the total no. of heads occuring in three tosses of an unbiased coin.

ans: Discrete random variable.

Let, X = no. of heads occuring.

$X = \{0, 1, 2, 3\}$ (0 head, 1 head, 2 heads, all heads)

3 tosses:

Total outcomes $= 2^3 = 8$ Head=1, Tail=0.

$$P(X=0) = P(TTT) = \frac{1}{8}$$

$$P(X=1) = P(HTT, THT, TTH) = \frac{3}{8}$$

$$P(X=2) = P(HHT, HTH, THH) = \frac{3}{8}$$

$$P(X=3) = P(HHH) = \frac{1}{8}$$

PMF / PDF is given by

X:	0	1	2	3
P(X):	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Probability function:

$$F(x) = P(X \leq x) = \Sigma P(x)$$

$X \leq 1 = 0, 1$ $\quad X \leq 2 = 0, 1, 2$

X:	0	1	2	3
F(x):	$\frac{1}{8}$	$\frac{1}{8}+\frac{3}{8}$ $= \frac{1}{2}$	$\frac{1}{8}+\frac{3}{8}+\frac{3}{8}=\frac{7}{8}$	1

Mean:

$$E(X) = \Sigma x P(x)$$

$$= \left(\frac{1}{8} \times 0\right) + \left(\frac{3}{8} \times 1\right) + \left(\frac{3}{8} \times 2\right) + \left(\frac{1}{8} \times 3\right)$$

$$E(X) = 0 + \frac{3}{8} + \frac{3}{4} + \frac{3}{8} = \frac{12}{8} = \frac{3}{2} = 1.5$$

Q: Let X be a number on a die when thrown. Find the mean and variance of X.

ans: $X = \{1, 2, 3, 4, 5, 6\}$

$$P(X=1) = \frac{1}{6}$$
$$P(X=2) = \frac{1}{6}$$
$$P(X=3) = \frac{1}{6}$$
$$P(X=4) = \frac{1}{6}$$
$$P(X=5) = \frac{1}{6}$$
$$P(X=6) = \frac{1}{6}$$

X:	1	2	3	4	5	6
P(X):	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Mean: $E(X) = \Sigma x P(x)$

$$= \frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6}$$

$$E(X) = \frac{21}{6} = \frac{7}{2}$$

$$Var(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum x^2 P(x)$$

$$= \left(1^2 \times \frac{1}{6}\right) + \left(2^2 \times \frac{1}{6}\right) + \left(3^2 \times \frac{1}{6}\right) + \left(4^2 \times \frac{1}{6}\right) + \left(5^2 \times \frac{1}{6}\right) + \left(6^2 \times \frac{1}{6}\right)$$

$$E(X^2) = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$$

$$Var(X) = \frac{91}{6} - \left(\frac{21}{6}\right)^2$$

$$= \frac{91}{6} - \frac{441}{36} = \frac{546-441}{36} = \frac{105}{36}$$

$$Var(X) = \frac{35}{12}$$

THEORETICAL DISTRIBUTION

Binomial / Bernoulli's distribution: (only two possible outcomes - success or failure).

p - probability of success

q - probability of failure.

$$p+q=1$$

n - no. of trials.

r - successes. - ${}^nC_r$ ways it can happen.

If x times success:

$$\therefore P(X=x) = {}^nC_x \, P(\underbrace{SSS\ldots S}_{x \text{ times}} \underbrace{F F \ldots F}_{n-x \text{ times}})$$

$$= {}^nC_x \, \underbrace{PPPP\ldots P}_{x \text{ times}} \cdot \underbrace{qqqq\ldots q}_{n-x \text{ times}}$$

$$P(X=x) = {}^nC_x \, p^x q^{n-x}, \quad x = 0, 1, \ldots, n$$

↑ PMF / PDF of Binomial distribution.

$x=0$, $P(X=0) = {}^nC_0 \, q^n$

$x=1$, $P(X=1) = {}^nC_1 \, pq^{n-1}$

$(q+p)^n$ - it is called binomial because ${}^nC_0, {}^nC_1 \ldots$ are the coefficients of the binomial expansion of $(q+p)^n$.

- n - should be finite.
- All the trials are independent of each other.
- If p is constant, then q is constant.

THEORETICAL DISTRIBUTIONS

- Frequency distributions can be classified under two heads:
  (i) observed frequency distributions.
  (ii) theoretical or expected frequency distributions.
- Observed frequency distributions are based on actual observations and experimentation.
- If certain hypothesis is assumed, it is sometimes possible to derive mathematically what the frequency distribution of certain universe should be such distributions are called Theoretical Distributions.
- There are many types of theoretical frequency distributions, but we consider the following:
  (i) Binomial Distribution (or Bernoulli's Distribution)
  (ii) Poisson's Distribution
  (iii) Normal Distribution.

1. Binomial (or Bernoulli's) Distribution

Binomial Probability Distribution

- Let there be n independent trials in an experiment.
- Let a r.v X denote the no. of successes in these n trials. Let p be the probability of a success and q that of a failure in a single trial so that $p+q=1$. Let the trials be independent and p be constant for every trial.
- The probability of r successes in n trials.
  'r' successes can be obtained in n trials in ${}^nC_r$ ways.

$$\therefore P(X=x) = {}^nC_x \, P(\underbrace{SSS\ldots\ldots S}_{x \text{ times}} \underbrace{FFF\ldots\ldots F}_{(n-x) \text{ times}})$$

$$= {}^nC_x \underbrace{P(S)\,P(S)\ldots\ldots P(S)}_{x \text{ factors}} \underbrace{P(F)\,P(F)\ldots\ldots P(F)}_{(n-x) \text{ factors}}$$

$$= {}^nC_x \, p^x q^{n-x} \quad \text{— (1)}$$

Hence $P(X=x) = {}^nC_x\, q^{n-x}\, p^x$, where $p+q=1$ and $x=0,1,\ldots,n$.

The distribution ① is called the binomial probability distribution and x is called the binomial variate.

Remark

1. $P(X=R)$ is usually written as $P(r)$.
2. The successive probabilities $P(r)$ in ① for $r=0,1,2,\ldots,n$ are
$${}^nC_0\, q^n,\ {}^nC_1\, q^{n-1}p,\ {}^nC_2 q^{n-2}p^2,\ \ldots\ldots,\ {}^nC_n\, p^n.$$
which are the successive terms of the binomial expansion of $(q+p)^n$. That is why this distribution is called binomial distribution.
3. n and p occurring in the binomial distribution are called the parameters of the distribution.
4. In a binomial distribution:
   (i) n, the number of trials in finite
   (ii) each trial has only two possible outcomes usually called success and failure.
   (iii) all the trials are independent.
   (iv) p (and hence q) is constant for all trials.

23-10-2024
V.V
Derivation ✗

MEAN & VARIANCE OF BINOMIAL DISTRIBUTION

For binomial distribution,

$$PDF = {}^nC_x\, p^x q^{n-x}$$

$$PDF,\ P(x) = \begin{cases} {}^nC_x\, p^x q^{n-x}, & x=0,1,\ldots,n \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Mean} = \mu = E(X) = \sum_{x=0}^{n} x P(x)$$

$$\mu = \sum_{x=0}^{n} x\, {}^nC_x p^x q^{n-x}$$

$$\mu = 0 + {}^nC_1 pq^{n-1} + 2\,{}^nC_2 p^2 q^{n-2} + \ldots\ldots + n\cdot {}^nC_n p^n$$

$$\mu = nq^{n-1}p + 2\cdot\frac{n(n-1)}{2\times1}q^{n-2}p^2 + \frac{3\cdot n(n-1)(n-2)}{3\times2\times1}q^{n-3}p^3 + \ldots + np^n.$$

$$= np\left(q^{n-1} + \frac{2(n-1)}{}q^{n-2}p + \frac{(n-1)(n-2)}{2\times1}q^{n-3}p^2 + \ldots\right)$$

$$= np\left({}^{n-1}C_0\,q^{n-1} + {}^{n-1}C_1\,q^{n-2}p + {}^{n-1}C_2\,q^{n-3}p^2 + \ldots\right)$$

$$= np(q+p)^{n-1}$$

$$= np(1)^{n-1}$$

$$\mu = np$$

Mean of binomial distribution, $\mu = np$.

Variance of $X = E(X^2) - (E(X))^2$

$$\sigma^2 = \text{var}(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum_{x=0}^{n} x^2 P(x)$$

$$= \sum_{x=0}^{n} x^2\ {}^nC_x\,p^x q^{n-x} \qquad \left(x^2 = x + x(x-1)\right)$$

$$= \sum_{x=0}^{n}\left[x + x(x-1)\right]{}^nC_x\,p^x q^{n-x}$$

$$= \sum_{x=0}^{n} x\,{}^nC_x\,p^x q^{n-x} + \sum_{x=0}^{n} x(x-1)\,{}^nC_x\,p^x q^{n-x}$$

$$= np + \sum_{x=0}^{n} x(x-1)\,{}^nC_x\,p^x q^{n-x} \qquad (\because \mu = np = \textstyle\sum x\,P(x))$$

$$= np + \left(0 + 0 + 2(1)\,{}^nC_2\,p^2q^{n-2} + 3(2)\,{}^nC_3\,q^{n-3}p^3 + \ldots\right.$$
$$\left. \ldots\ldots + n(n-1)\,{}^nC_n\,p^n\right)$$

$$= np + \left(2\,\frac{n(n-1)}{2\times1}\,p^2q^{n-2} + 6\,\frac{n(n-1)(n-2)}{3\times2\times1}\,q^{n-3}p^3 + \ldots\right.$$
$$\left. \ldots + n(n-1)p^n\right)$$

$$= np + n(n-1)p^2 \left[q^{n-2} + (n-2)q^{n-3}p + \cdots + p^{n-2}\right]$$

$$= np + n(n-1)p^2 \left[{}^{n-2}C_0\, q^{n-2} + {}^{n-2}C_1\, q^{n-3}p + \cdots + {}^{n-2}C_{n-2}\, p^{n-2}\right]$$

$$= np + n(n-1)p^2 (q+p)^{n-2}$$

$$E(X^2) = np + n(n-1)p^2 \qquad (\because q+p=1)$$

$$\text{Var}(X) = \sigma^2 = E(X^2) - [E(X)]^2$$

$$= np + n(n-1)p^2 - (np)^2$$

$$= np + n^2p^2 - np^2 - n^2p^2$$

$$= np - np^2$$

$$= np(1-p)$$

$$\text{Var}(X) = npq$$

Variance of binomial distribution, $\text{var}(X) = npq$.

Standard deviation of binomial distribution, $\sigma = \sqrt{npq}$.

Q: Mean and variance of binomial variate X are 16 and 8, find (i) $P(X=0)$

(ii) $P(X \geq 2)$.

ans: Mean, $\mu = np = 16$

$\sigma^2 = npq = 8$

$\therefore q = \frac{1}{2}$

$(\because p+q=1)$

$np = 16$

$p = 1 - \frac{1}{2} = \frac{1}{2}$

$n = 16 \times 2 = 32$

(i) $n = 32, \quad x = 0$

$f(x) = {}^nC_x p^x q^{n-x}$

$P(x=0) = {}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0}$

$f(x=0) = \frac{1}{2^{32}}$

(ii) $P(X \geq 2) = 1 - P(X \leq 1) = 1 - [f(0) + f(1)]$ ($P(X \leq 2) = P($ $P($

$= 1 - \left[{}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0} + {}^{32}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{32-1} \right]$

$= 1 - \left[\frac{1}{2^{32}} + 32 \times \frac{1}{2^{32}} \right]$

$P(x \geq 2) = 1 - \frac{33}{2^{32}} = 0.9999992 \approx 1$

29-10-2024

Q: Find the binomial distribution, whose mean is 3 and variance is 2.

ans:

Mean $= np = 3$

Variance $= npq = 2$

$\therefore q = \frac{2}{3}$

$p = 1 - q = 1 - \frac{2}{3} = \frac{1}{3}$

$n = \frac{2}{\frac{2}{9}} = 9$

Binomial distribution $= (p+q)^n$ or ${}^nC_x p^x q^{n-x}$

$\therefore$ Binomial distribution $= \left(\frac{2}{3} + \frac{1}{3}\right)^9$ or ${}^9C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{9-x}$

8 10 coins are [illegible] probability of [illegible] atleast 7 heads. [illegible] probability

ans Probability of getting head = $p = \frac{1}{2}$

$q = \frac{1}{2}$

$n = 10$

$P(x) = {}^nC_x\, p^x q^{n-x}$

Probability of getting atleast seven heads,

$P(\text{getting atleast 7 heads}) = P(X \geq 7)$

$= P(10) + P(9) + P(8) + P(7)$

$= {}^{10}C_{10}\left(\frac{1}{2}\right)^{10}\left(\frac{1}{2}\right)^{10-10} + {}^{10}C_9\left(\frac{1}{2}\right)^9\left(\frac{1}{2}\right)^{10-9} + {}^{10}C_8\left(\frac{1}{2}\right)^8\left(\frac{1}{2}\right)^{10-8}$

$+ {}^{10}C_7\left(\frac{1}{2}\right)^7\left(\frac{1}{2}\right)^{10-7}$

$= \frac{1}{2^{10}} + 10 \times \frac{1}{2^{10}} + \frac{10 \times 9}{2}\,\frac{1}{2^{10}} + \frac{10 \times 9 \times 8}{3!} \times \frac{1}{2^{10}}$

$= \frac{1 + 10 + 45 + 120}{2^{10}} = \frac{176}{1024} = \frac{11}{64}$

Q: During war, one ship out of nine was sunk on an average in making a certain voyage. What was the probability that exactly three out of a convoy of 6 ships would arrive safely?

ans: Let X be the no. of ships of a convoy of 6 which completes the voyage safely.

One of nine sunk, $\therefore q = \frac{1}{9}$.

$\therefore p = \frac{8}{9}$.

$n = 6$.

P(probability of three out of 6 reached safely)

$= P(X = 3)$

(i) $n=32, \quad x=0$

$f(x) = {}^nC_x p^x q^{n-x}$

$P(x=0) = {}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0}$

$P(x=0) = \frac{1}{2^{32}}$

(ii) $P(X \geq 2) = 1 - P(X \leq 1) = 1 - [f(0) + f(1)]$ ($P(X \leq 2) = P(0) + P(1) + P(2)$)

$= 1 - \left[{}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0} + {}^{32}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{32-1} \right]$

$= 1 - \left[\frac{1}{2^{32}} + 32 \times \frac{1}{2^{32}} \right]$

$P(X \geq 2) = 1 - \frac{33}{2^{32}} = 0.\overline{999999}2 \approx 1$

29-10-2024

Q: Find the binomial distribution, whose mean is 3 and variance is 2.

ans:

Mean $= np = 3$

Variance $= npq = 2$

$\therefore q = \frac{2}{3}$

$p = 1 - q = 1 - \frac{2}{3} = \frac{1}{3}$

$n = \frac{2}{\frac{2}{9}} = 9$

Binomial distribution $= (p+q)^n$ or ${}^nC_x p^x q^{n-x}$

$\therefore$ Binomial distribution $= \left(\frac{2}{3} + \frac{1}{3}\right)^9$ or ${}^9C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{9-x}$

probability of getting atleast 7 heads.

ans

Probability of getting head $= p = \frac{1}{2}$

$q = \frac{1}{2}$

$n = 10$

$P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$

Probability of getting atleast seven heads,

$P(\text{getting atleast 7 heads}) = P(X \geq 7)$

$= P(10) + P(9) + P(8) + P(7)$

$= {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10-10} + {}^{10}C_{9} \left(\frac{1}{2}\right)^{9} \left(\frac{1}{2}\right)^{10-9} + {}^{10}C_{8} \left(\frac{1}{2}\right)^{8} ($

$+ {}^{10}C_{7} \left(\frac{1}{2}\right)^{7} \left(\frac{1}{2}\right)^{10-7}$

$= \frac{1}{2^{10}} + 10 \times \frac{1}{2^{10}} + \frac{10 \times 9}{2} \frac{1}{2^{10}} + \frac{10 \times 9 \times 8}{3!} \times$

$= \frac{1 + 10 + 45 + 120}{2^{10}} = \frac{176}{1024} = \frac{11}{64}$

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$P(\text{probability of three out of 6 reached safely}) = P(X = 3)$

(i) $n=32, \quad x=0$

$$f(x) = {}^nC_x p^x q^{n-x}$$

$$P(x=0) = {}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0}$$

$$P(x=0) = \frac{1}{2^{32}}$$

(ii) $P(X \geq 2) = 1 - P(X \leq 1) = 1 - [f(0) + f(1)]$ $\quad (\because P(X \leq 2) = P(0) + P(1) + P($

$$= 1 - \left[{}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0} + {}^{32}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{32-1} \right]$$

$$= 1 - \left[\frac{1}{2^{32}} + 32 \times \frac{1}{2^{32}} \right]$$

$$P(x \geq 2) = 1 - \frac{33}{2^{32}} = 0.9999992 \approx 1$$

29-10-2024

Q: Find the binomial distribution, whose mean is 3 and variance is 2.

ans:

Mean $= np = 3$

Variance $= npq = 2$

$\therefore q = \frac{2}{3}$

$p = 1 - q = 1 - \frac{2}{3} = \frac{1}{3}$

$n = \frac{2}{\frac{2}{9}} = 9$

Binomial distribution $= (p+q)^n$ or ${}^nC_x p^x q^{n-x}$

$\therefore$ Binomial distribution $= \left(\frac{2}{3} + \frac{1}{3}\right)^9$ or ${}^9C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{9-x}$

Q. 10 coins are thrown simultaneously, find the probability of getting atleast 7 heads.

ans

Probability of getting head $= p = \frac{1}{2}$

$q = \frac{1}{2}$.

$n = 10$

$P(x) = {}^{n}C_x \, p^x q^{n-x}$

Probability of getting atleast seven heads,

P(getting atleast 7 heads) $= P(X \geq 7)$

$= P(10) + P(9) + P(8) + P(7)$

$= {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10-10} + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{10-9} + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8}$

$+ {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{10-7}$

$\frac{10!}{3! \times 2!}$

$= \frac{1}{2^{10}} + 10 \times \frac{1}{2^{10}} + \frac{10 \times 9}{2} \frac{1}{2^{10}} + \frac{10 \times 9 \times 8}{3!} \times \frac{1}{2^{10}}$

$= \frac{1 + 10 + 45 + 120}{2^{10}} = \frac{176}{1024} = \frac{11}{64}$

Q: During war, one ship out of nine was sunk on an average in making a certain voyage. What was the probability that exactly three out of a convoy of 6 ships would arrive safely?

ans: Let x be the no. of ships of a convoy of 6 which completes the voyage safely.

One of nine sunk, $\therefore q = \frac{1}{9}$.

$\therefore p = \frac{8}{9}$.

$n = 6$.

P(probability of three out of 6 reached safely)

$= P(X = 3)$

PDF = $f(x) = {}^nC_x p^x q^{n-x}$

(i) $n = 32, \quad X = 0.$

$$f(x) = {}^nC_x p^x q^{n-x}$$

$$P(X=0) = {}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0}$$

$$P(X=0) = \frac{1}{2^{32}}$$

(ii) $P(X \geq 2) = 1 - P(X \leq 1) = 1 - [f(0) + f(1)]$ $(\because P(X \leq 2) = P(0) + P(1))$

$$= 1 - \left[{}^{32}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{32-0} + {}^{32}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{32-1} \right]$$

$$= 1 - \left[\frac{1}{2^{32}} + 32 \times \frac{1}{2^{32}} \right]$$

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29-10-2024

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Q: 10 coins are thrown simultaneously, find the ~~probabilbi~~ probability of getting atleast 7 heads.

ans: Probability of getting head $= p = \frac{1}{2}$.

$q = \frac{1}{2}$.

$n = 10$

$P(X) = {}^nC_x\, p^x q^{n-x}$

Probability of getting atleast seven heads,

P(getting atleast 7 heads) $= P(X \geq 7)$

$= P(10) + P(9) + P(8) + P(7)$

$= {}^{10}C_{10}\left(\frac{1}{2}\right)^{10}\left(\frac{1}{2}\right)^{10-10} + {}^{10}C_9\left(\frac{1}{2}\right)^9\left(\frac{1}{2}\right)^{10-9} + {}^{10}C_8\left(\frac{1}{2}\right)^8\left(\frac{1}{2}\right)^{10-8}$

$+ {}^{10}C_7\left(\frac{1}{2}\right)^7\left(\frac{1}{2}\right)^{10-7}$

$\frac{10!}{3! \times 2!}$

$= \frac{1}{2^{10}} + 10 \times \frac{1}{2^{10}} + \frac{10 \times 9}{2}\frac{1}{2^{10}} + \frac{10 \times 9 \times 8}{3!} \times \frac{1}{2^{10}}$

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Q: During war, one ship out of nine was sunk on an average in making a certain voyage. What was the probability that exactly three out of a convoy of 6 ships would arrive safely?

ans: Let X be the no. of ships of a convoy of 6 which completes the voyage safely.

One of nine sunk, $\therefore q = \frac{1}{9}$.

$\therefore p = \frac{8}{9}$.

$n = 6$.

P(probability of three out of 6 reached safely)

$= P(X = 3)$

$P(X) = {}^nC_x p^x q^{n-x}$

$P(X=3) = {}^6C_3 \left(\frac{8}{9}\right)^3 \left(\frac{1}{9}\right)^{6-3}$

$= {}^6C_3 \times \frac{8^3}{9^3} \times \frac{1}{9^3}$ $= \frac{20 \times 512}{531441}$

$P(X=3) = 0.01927$

FITTING OF A BINOMIAL DISTRIBUTION TO AN OBSERVED FREQUENCY DISTRIBUTION

- The process of determining the most appropriate value of the parameters from the given observations and writing down the PDF is known as fitting of the distribution.
- Let a frequency table be given such that X takes value $0, 1, 2, \ldots n$ and the corresponding frequencies $f_0, f_1, \ldots f_n$. Then $\sum_{i=1}^{n} f_i = N$.
- It is assumed that X follows a binomial distribution ${}^nC_x p^x q^{n-x}$ and if the total no of observations $= N$ we expect that $X=x$ in $N\,{}^nC_x p^x q^{n-x}$ cases! This is called the theoretical frequency (or expected frequency of $X=x$.

Q: The following data are the no of seeds germinating out of 10 on a damp filter for 80 sets of seeds. Fit a binomial distribution to this data.

$X=x$	0	1	2	3	4	5	6	7	8	9	10
f (Frequency)	6	20	28	12	8	6	0	0	0	0	0

ans: $N = \sum_{i=1}^{n} f_i = 80$

$n = 10$

$$\bar{x} = \frac{\Sigma f x}{\Sigma f}$$ Mean of

Mean of observed distribution.

$$\bar{x} = \frac{1}{80}[(0\times 6) + (1\times 20) + (2\times 28) + (3\times 12) + (4\times 8) + (5\times 6) + 0 + 0 + 0 + 0 + 0]$$

$$\bar{x} = 2.175$$

$$\bar{x} = np$$

$$np = 2.175$$

$$p = \frac{2.175}{10} = 0.2175$$

$$q = 1 - p = 0.7825$$

Theoretical frequency $= N\ {}^{n}C_x\ p^x q^{n-x}$

$= 80\ {}^{10}C_x\ p^x\ q^{10-x}$

$= 80\ {}^{10}C_x\ (0.2175)^x\ (0.7825)^{10-x}$

PDF $= (q+p)^n$

x	0	1	2

For $x=0$, $80\ {}^{10}C_0\ (0.2175)^0 (0.7825)^{10} = 6.885$

$x=1$, $80\ {}^{10}C_1\ (0.2175)^1 (0.7825)^9 = 19.1385$

$x=2$, $80\ {}^{10}C_2\ (0.2175)^2 (0.7825)^8 = 23.938$

$x=3$, $80\ {}^{10}C_3\ (0.2175)^3 (0.7825)^7 = 17.743$

$x=4$, $80\ {}^{10}C_4\ (0.2175)^4 (0.7825)^6 = 8.6308$

$x=5$, $80\ {}^{10}C_5\ (0.2175)^5 (0.7825)^5 = 2.8788$

$x=6$, $80\ {}^{10}C_6\ (0.2175)^6 (0.7825)^4 = 0.6668$

$x=7$, $80\ {}^{10}C_7\ (0.2175)^7 (0.7825)^3 = 0.1059$

$x=8$, $80\ {}^{10}C_8\ (0.2175)^8 (0.7825)^2 = 0.01104$

$x=9$, $80\ {}^{10}C_9\ (0.2175)^9 (0.7825)^1 = 0.00068$

$x=10$, $80\ {}^{10}C_{10}\ (0.2175)^{10} (0.7825)^0 = 0.000018953$

x:	0	1	2	3	4	5	6	7	8	9	10
Theoretical f:	6.885	19.1385	23.938	17.743	8.6308	2.8788	0.6668	0.1059	0.01104	0.00068	0.000018953

Q: Five die were thrown together 96 times the no. of tim
4 or 5 or 6 was actualy thrown in the experiment
is given below. Calculate the expected frequencies.

No of die showing 4,5 or 6 :	0	1	2	3	4	5
Observed frequency :	1	10	24	35	18	8

Ans:

$P(\text{getting } 4,5 \text{ or } 6) = p = \frac{3}{6} = \frac{1}{2}$

$\therefore q = \frac{1}{2}$

$n = 5 \quad (0,1,2,\ldots,n)$

$N = 96$

Binomial distribution $= \left(\frac{1}{2} + \frac{1}{2}\right)^5$

Theoretical frequency $= N\,{}^nC_x\, p^x q^{n-x}$

$= 96\,{}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{n-x}$

For

$x=0, \quad 96\,{}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{96}{2^5} = \frac{96}{32} = 3$

$x=1, \quad 96\,{}^5C_1 \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)^{5-1} = \frac{96 \times 5}{32} = 15$

$x=2, \quad 96\,{}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2} = 96 \times \frac{5 \times 4}{2} \times \frac{1}{32} = 30$

$x=3, \quad 96\,{}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3} = 96 \times \frac{5 \times 4}{2} \times \frac{1}{32} = 30$

$x=4, \quad 96\,{}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{5-4} = 96 \times 5 \times \frac{1}{32} = 15$

$x=5, \quad 96\,{}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{5-5} = \frac{96}{32} = 3.$

No. of die showing 4,5 or 6 :	0	1	2	3	4	5
Theoretical freq :	3	15	30	30	15	3

POISSON DISTRIBUTION

A random variable X is said to follow the Poisson distribution if its PDF is

$$f(x) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!}, & x = 0, 1, 2, \ldots \quad (\text{to } \infty) \\ 0, & \text{otherwise} \end{cases} \quad (\lambda \geq 0)$$

~~$(\lambda \geq 0)$~~

where λ is the parameter of distribution.

Sum of probabilities = 1.

Poisson distribution ~~has~~ a limiting case of binomial distribution.

Poisson Distribution as a limiting case of binomial Distribution:

Binomial distribution tends to or approaches poisson distribution under the following conditions:

(if n is very large and p is very small choose poisson)

(i) n, no. of trails is very large, $n \to \infty$

(ii) p, probability of success is very small, $p \to 0$

(iii) $np = \lambda$ (say) is finite.

$$np = \lambda$$

$$p = \frac{\lambda}{n} \quad \text{and} \quad q = 1 - \frac{\lambda}{n}$$

For binomial distribution,

$$f(x) = {}^nC_x \, p^x q^{n-x}, \quad x = 0, 1, 2, \ldots, n$$

$$= \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n(n-1)(n-2)\cdots(n-(x-1))}{x!} \times \left(\frac{\lambda}{n}\right)^x \times \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

(taking n, x times outside)

$$= \frac{n^x \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{x-1}{n}\right) \times \lambda^x \left(1 - \frac{\lambda}{n}\right)^n}{x! \times n^x \times \left(1 - \frac{\lambda}{n}\right)^x}$$

$$= \frac{1\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) \cdots \left(1-\frac{(x-1)}{n}\right) \lambda^x \left(1-\frac{\lambda}{n}\right)^n}{x! \quad \left(1-\frac{\lambda}{n}\right)^x}$$

When $n \to \infty$,

$$\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) \cdots \left(1-\left(\frac{x-1}{n}\right)\right) \longrightarrow 1$$

$$\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^x = 1$$

$$\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n = e^{-\lambda}$$

$$\therefore f(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, \ldots\ldots$$

This being the PDF of Poisson distribution under the above conditions.

MEAN & VARIANCE OF POISSON DISTRIBUTION

For Poisson distribution,

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$\text{Mean} = \mu = E(x) = \sum_{x=0}^{\infty} x f(x)$$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$$

$(0-1)!$ not possible, so $x = 0 \to \infty$ not possible

$$= e^{-\lambda}\left[\lambda + \frac{\lambda^2}{1!} + \frac{\lambda^3}{2!} + \ldots\ldots\right]$$

$$= e^{-\lambda}\lambda\left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \ldots\ldots\right]$$

$= e^{-\lambda} \lambda e^{\lambda} = \underline{\lambda}$

Mean $= \mu = \underline{\lambda}$

Variance $= \sigma^2 = E(X^2) - [E(X)]^2$

$$E(X^2) = \sum_{x=0}^{\infty} x^2 f(x) = \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda} \lambda^x}{x!}$$

$\rightarrow x$

$$= \sum_{x=0}^{\infty} (x + x(x-1)) \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} + \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \lambda + e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!}$$

$x(x-1) < \frac{1}{x(x-1)(x-2)!}$

$$= \lambda + e^{-\lambda} \left[\lambda^2 + \frac{\lambda^3}{1!} + \frac{\lambda^4}{2!} + \cdots \right]$$

$$= \lambda + e^{-\lambda} \lambda^2 \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots \right]$$

$$= \lambda + e^{-\lambda} \lambda^2 e^{\lambda}$$

$$E(X^2) = \lambda + \lambda^2$$

$\therefore$ Variance $= \sigma^2 = \lambda + \lambda^2 - (\lambda)^2 = \underline{\lambda}$

$\therefore$ Variance $= \sigma^2 = \underline{\lambda}$

Q: If x follows the Poisson distribution $P(X=1) = P(X=2)$. Then find $P(X=4)$.

ans:

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X=1) = \frac{e^{-\lambda} \lambda^1}{1!} = e^{-\lambda} \lambda$$

$$P(X=2) = \frac{e^{-\lambda} \lambda^2}{2!}$$

$$P(X=1) = P(X=2)$$

$$e^{-\lambda}\lambda = \frac{e^{-\lambda}\lambda^2}{2}$$

$$\Rightarrow \quad \lambda = 2$$

$$\therefore P(X=4) = \frac{e^{-2}\,2^4}{4!} = 0.09022$$

Q: If x is a Poisson variate such that $P(X=2) = 9P(X=4) + 90P(X=6)$. Find mean and standard deviation.

ans:

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$P(X=2) = \frac{e^{-\lambda}\lambda^2}{2!}$$

$$P(X=4) = \frac{e^{-\lambda}\lambda^4}{4!}$$

$$P(X=6) = \frac{e^{-\lambda}\lambda^6}{6!}$$

$$P(X=2) = 9P(X=4) + 90P(X=6)$$

$$\frac{e^{-\lambda}\lambda^2}{2!} = 9 \times \frac{e^{-\lambda}\lambda^4}{4!} + 90 \times \frac{e^{-\lambda}\lambda^6}{6!}$$

$$\frac{1}{2} = \frac{9\lambda^2}{24} + \frac{90\times\lambda^4}{720}$$

$$\Rightarrow \frac{1}{2} = \frac{3\lambda^2}{8} + \frac{3\lambda^4}{28}$$

$$\Rightarrow \frac{1}{2} = \frac{3\lambda^2+\lambda^4}{84}$$

$$\Rightarrow \lambda^4 + 3\lambda^2 = 4$$

$$\lambda^4 + 3\lambda^2 - 4 = 0$$

$$(\lambda^2+4)(\lambda^2-1) = 0$$

$$\lambda^2 = -4, 1$$

$\lambda^2 = 1$ (as λ^2 cannot be -ve)

Mean $= \mu = \lambda = \underline{1}$

Variance $= \sigma^2 = \lambda = \underline{1}$

Standard deviation $= \underline{1}$ $(\sqrt{\lambda})$

Q: Assume that the probability of an individual coal miner being killed in a mine accident during an year $\frac{1}{2400}$. Use Poisson distribution to calculate the probability that in a mine employing 200 miners, there will be atleast one fatal accident in an year.

ans:

$$p = \frac{1}{2400}$$

$$n = 200$$

$$\lambda = np = 200 \times \frac{1}{2400} = \frac{1}{12} = 0.0833$$

P(proba of atleast 1 fatal accident)

$$= 1 - P(\text{no fatal accident})$$

$$= 1 - \frac{e^{-\lambda}\lambda^x}{x!}$$

$$P(\text{atleast 1 fatal accident}) = 1 - \frac{e^{-0.0833}\lambda^0}{0!} = 0.0799$$

HOMEWORK

1. X follows a B.D with parameters $n=6$ and p. If $9P(X=4) = P(X=2)$ find p.

ans

$9P(X=4) = P(X=2)$ $\quad n=6$

$P(X=x) = {}^nC_x p^x q^{n-x}$

$P(X=4) = {}^6C_4\, p^4 q^{6-4} = 15p^4q^2$

$P(X=2) = {}^6C_2\, p^2 q^{6-2} = 15p^2q^4$

$9 \times 15 p^4 q^2 = 15 p^2 q^4$

$9p^2 = q^2$

$\Rightarrow 9p^2 = (1-p)^2 \Rightarrow 9p^2 = 1 - 2p + p^2$

$\Rightarrow 8p^2 + 2p - 1 = 0$

$p = 0.25, -0.5$

$\therefore p = 0.25$

2. Eight coins are thrown simultaneously. Find the chance of obtaining at least 6 heads.

ans:

$p = \frac{1}{2}$

$q = \frac{1}{2}$

$n = 8$

$P(X) = {}^nC_x p^x q^{n-x} = {}^8C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{n-x}$

P(getting atleast 6 heads) $= P(X \geq 6)$

$= P(X=8) + P(X=7) + P(X=6)$

$= {}^8C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^0 + {}^8C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^1 + {}^8C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^2$

$= \frac{1}{2^8} + 8 \times \frac{1}{2^8} + 28 \times \frac{1}{2^8}$

$= \frac{37}{2^8} = \frac{37}{256}$

3. The probability of any ship of a company being destroyed on a certain voyage is 0.02. The company owns 6 ships for the voyage. What is the probability of (i) losing one ship (ii) losing atmost two ships (iii) losing none?

ans.

$q = 0.02$

$P = 0.98$

$n = 6$

$P(X) = {}^{n}C_x \, p^x q^{n-x}$

(i) Losing one ship.

$P(X=1) = {}^{6}C_1 (0.98)^1 \times (0.02)^{6-1}$

$= 6 \times 0.98 \times 3.2 \times 10^{-9}$

$P(X=1) = 1.8816 \times 10^{-8}$

(ii) Losing atmost two ships.

$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= {}^{6}C_0 (0.98)^0 (0.02)^6 + {}^{6}C_1 (0.98)^1 (0.02)^5 + {}^{6}C_2 (0.98)^2 (0.02)^4$

$P(X \leq 2) = 2.32384 \times 10^{-6}$

(iii) Losing none.

$P(X=0) = {}^{6}C_0 (0.98)^0 (0.02)$

$P = 0.02$ (losing)

$q = 0.98$

$n = 6$

$P(X) = {}^{n}C_x \, p^x q^{n-x}$

(i) Losing one ship.

$P(X=1) = {}^{6}C_1 (0.02)^1 \times (0.98)^5 = 0.1085$

(ii) Losing atmost two ships.

$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= {}^{6}C_0 (0.02)^0 (0.98)^6 + {}^{6}C_1 (0.02)^1 (0.98)^5 + {}^{6}C_2 (0.02)^2 (0.98)^4$

$= 0.9998$

(iii) Losing none.

$P(X=0) = {}^{6}C_0 (0.02)^0 (0.98)^6 = 0.8858$

4 A set of 8 symmetrical coins was tossed 256 times and the frequencies of throws observed were as follows:

No. of heads:	0	1	2	3	4	5	6	7	8
Freq. of throws:	2	6	24	63	64	50	36	10	1

Fit a B.D to the data.

ans:

Mean, $\bar{x} = \frac{1}{256}(0+6+48+189+256+250+216+70+8)$

$\bar{x} = 4.0742$

$n = 8$

$N = 256$

$np = 4.0742$

$p = 0.5093$

$q = 1-p = 0.4907$

Theoretical frequency $= N\,{}^nC_x\,p^x q^{n-x}$

$= 256\,{}^8C_x\,(0.5093)^x\,(0.4907)^{8-x}$

For $x=0$, $TF = 256\,{}^8C_0(0.5093)^0(0.4907)^8 = 0.86$

$x=1$, $TF = 256\,{}^8C_1(0.5093)^1(0.4907)^7 = 7.145$

$x=2$, $TF = 256\,{}^8C_2(0.5093)^2(0.4907)^6 = 25.952$

$x=3$, $TF = 256\,{}^8C_3(0.5093)^3(0.4907)^5 = 53.88$

$x=4$, $TF = 256\,{}^8C_4(0.5093)^4(0.4907)^4 = 69.9$

$x=5$, $TF = 256\,{}^8C_5(0.5093)^5(0.4907)^3 = 58$

$x=6$, $TF = 256\,{}^8C_6(0.5093)^6(0.4907)^2 = 30.121$

$x=7$, $TF = 256\,{}^8C_7(0.5093)^7(0.4907)^1 = 8.932$

$x=8$, $TF = 256\,{}^8C_8(0.5093)^8(0.4907)^0 = 1.159$

5. If the probability that an item produced by a certain machine will be defective is 0.01, Show that the probability that a random sample of 100 items selected from the total output contains not more than one defective item is $\frac{2}{e}$.

Soln:

$P = 0.01$

$n = 100$

$\lambda = np = 0.01 \times 100 = 1$

$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$

P(not more than one defective) $= P(X \leq 1)$

$= P(X=1) + P(X=0)$

$= \frac{e^{-1}\,1^1}{1!} + \frac{e^{-1}\,1^0}{0!}$

$= e^{-1} + e^{-1} = 2e^{-1} = \frac{2}{e}$

05-10-2024

NORMAL DISTRIBUTION (N.D)

- continuous distribution.
- Normal distribution is continuous distribution it can be derived from the B.D in the limiting cased, where (i) n, no. of trials is very large & (ii) p, probability of success is close to ½.
- A continuous r.v X, with PDF,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

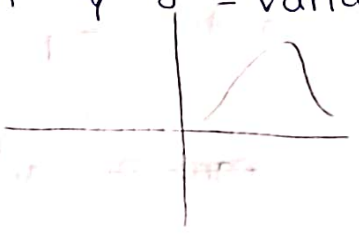
is said to follow normal distribution with parameters (μ, σ) and is denoted by $N(\mu, \sigma^2)$.

r.v X follows Normal distribution N.

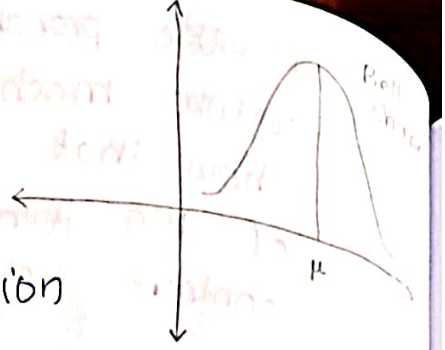
μ - mean, σ - is the std deviation & σ^2 variance.

$-\infty < \mu < \infty$

and $\sigma > 0$.



Properties



1. ND is a continuous distribution.
2. The mean of the distribution is μ, Variance is σ^2 and standard deviation is σ
3. Normal curve is bell shaped and it is symmetrical about the line $x=\mu$.
4. $\int_{-\infty}^{\infty} f(x)\,dx = 1$. (Total area under the normal curve=1).
5. Since, $f(x) \geq 0$, the curve always lies above the X-axis. The curve contains the maximum probability, at $x=\mu$

$$[f(x)]_{max} = \frac{1}{\sigma\sqrt{2\pi}}$$

6. As σ increases, the curve flattens out and as σ decreases, f value increases.
7. $P(a \leq x \leq b) = \int_a^b f(x)\,dx$.
8. It is a unimodel distribution. (The Mean, median, mode are same for a unimodel distribution.).

Standard Normal Distribution (Standard Form of ND)

- If X is a normal random variable with mean μ and standard deviation σ, then

$$z = \frac{x-\mu}{\sigma}$$ has the ND with mean = 0 (μ=0) and $\sigma = 1$. (SD=1).

- The random variable, z is called the standard ~~random~~ normal variable.

The PDF for the standard normal distribution is given by $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$. $(\sigma = 1)$

This eqn is free of parameters.

NOTE:

1. If $f(z)$ is the probability density function for the N.D, then

$$P(z_1 < Z < z_2) = \int_{z_1}^{z_2} f(z)\, dz = F(z_2) - F(z_1)$$

(Distribution fn - F)

$$F(z) = \int_{-\infty}^{z} f(z)\, dz = P(Z \leq z)$$

The fn $F(z)$ is called the distribution function.

2. $P(z_1 \leq z \leq z_2)$, $P(z_1 \leq z < z_2)$, $P(z_1 < z \leq z_2)$ and $P(z_1 < z < z_2)$ are same.

3. $F(-z_1) = 1 - F(z_1)$.

MEAN & VARIANCE OF N.D

$$\text{Mean} = E(x) = \int_{-\infty}^{\infty} x f(x)\, dx$$

$$= \int_{-\infty}^{\infty} (x - \mu + \mu) f(x)\, dx$$

$$= \int_{-\infty}^{\infty} (x-\mu) f(x)\, dx + \int_{-\infty}^{\infty} \mu f(x)\, dx$$

$$= \int_{-\infty}^{\infty} (x-\mu) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx + \mu \int_{-\infty}^{\infty} f(x)\, dx$$

$$= 0 + \mu \times (1) \qquad \left(\because \int_{-\infty}^{\infty} (x-\mu) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \text{ is odd fn of } (x-\mu), \int \text{odd} = 0 \right.$$

$$\left. \& \int_{-\infty}^{\infty} f(x)\, dx = 1 \right)$$

$$= \mu$$

$\therefore$ Mean $= \mu = E(X)$

Variance of N.D

$$V(X) = E(X-\mu)^2$$

$$V(X) = E(X-\mu)^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$$

$$= \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma z)^2 e^{\frac{-1}{2}z^2} \sigma\, dz$$

[Put $\frac{x-\mu}{\sigma} = z$, $x-\mu = \sigma z$, $dx = \sigma dz$, $x = \pm\infty$, $z = \pm\infty$]

$$= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{\frac{-1}{2}z^2} dz = \frac{2\sigma^2}{\sqrt{2\pi}} \int_{0}^{\infty} z^2 e^{\frac{-1}{2}z^2} dz$$ (even fn)

Substitution, $\frac{z^2}{2} = u$,

i.e.

$$z^2 = 2u$$

$$2z\,dz = 2du$$

$$dz = \frac{du}{\sqrt{2u}}$$

$z = 0$, $u = 0$

$z = \infty$, $u = \infty$.

$$\therefore V(X) = \frac{2\sigma^2}{\sqrt{2\pi}} \int_{0}^{\infty} 2u^{\frac{1}{2}} e^{-u} \frac{du}{\sqrt{2u}}$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_{0}^{\infty} \sqrt{u}\, e^{-u} du$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_{0}^{\infty} u^{1/2} e^{-u} du$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_{0}^{\infty} e^{-u} u^{\frac{3}{2}-1} du$$

($\because \Gamma p = \int_0^\infty e^{-x} x^{p-1} dx$

$\Gamma p = (p-1)\Gamma(p-1)$

$\Gamma\frac{1}{2} = \sqrt{\pi}$)

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^\infty u^{3/2-1} e^{-u} du$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \left[\frac{3}{2} = \frac{2\sigma^2}{\sqrt{\pi}} \frac{1}{2} \left(\frac{1}{2}\right)\right] \quad (\because \lceil p = (p-1)\lceil p-1)$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \times \frac{1}{2} \times \sqrt{\pi} \quad (\because \left[\frac{1}{2} = \sqrt{\pi}\right)$$

$$\therefore \quad V(X) = \sigma^2$$

Q: A sample of 100 dry battery cells tested to find the length of life produces the following ~~cells~~ results,

$\bar{x} = 12$ hrs, $\sigma = 3$ hrs. Assuming the data to be normally distributed, what percentage of battery cells are expected to have life:

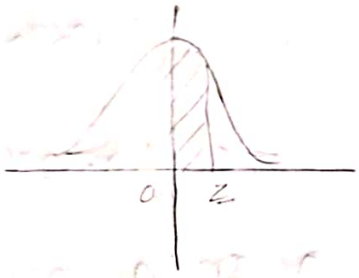
(i) more than 15 hours,

(ii) less than 6 hours, and

(iii) between 10 & 14 hours?

ans: Let x denotes the length of life of dry battery cells.

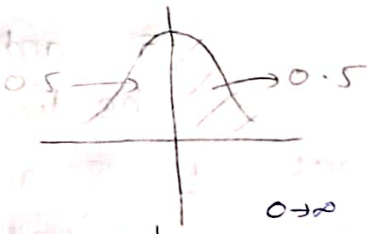
$$z = \frac{x - \bar{x}}{\sigma} = \frac{x-12}{3}$$



(i) $P(x > 15)$

when $x = 15$, $z = \frac{15-12}{3} = 1$.

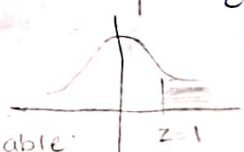
$P(x > 15) = P(z > 1)$



$= P(0 < z < \infty) - P(0 < z < 1)$

$= 0.5 - 0.3413$

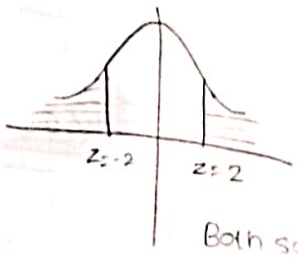
$P(x > 15) = 0.1587 = 15.87\%$



15.87% batteries have life >15 hrs.

Row	col
1.1	1.1 + 0
1.11	1.11 + 1
1.15	1.1 + 5
1.01	1.0 + 7

(ii) $x=6$, $z=\frac{x-12}{3}=\frac{6-12}{3}=-2$



Both same

$P(x<6) = P(z<-2)$

$= P(z>2)$

$= P(0<z<\infty) - P(0<z<2)$

$= 0.5 - 0.4772$

$P(x<6) = 0.0228 = 2.28\%$

2.28% batteries have lifespan < 6hrs.

(iii) $P(10<x<14)$

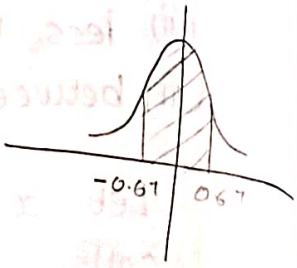
Here when, $x=10$, $z=\frac{-2}{3}=-0.67$

$x=14$, $z=\frac{2}{3}=0.67$

$P(10<x<14) = P(-0.67<z<0.67)$

$= 2\,P(0<z<0.67)$

$= 2 \times 0.248 = 0.4972$

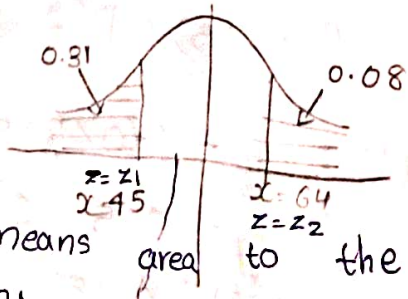


$P(10<x<14) = 49.72\%$

49.72% batteries have life span b/t 10 & 14 hrs.

Q: In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean & SD of the distribution.

ans: Let $\bar{x}$ and σ be the mean & standard deviation.



31% of items are under 45, means area to the left of the ordinate $x=45$ is 0.31.

When $x=45$, let $z=z_1$.

$P(z_1<z<0) = 0.5 - 0.31$

$P(z_1<z<0) = 0.19$ //

from the table the value corresponding to 0.19,

$z_1 = -0.5$ (-ve $\Rightarrow$ as it is left side of distribution).

When $x = 64$, Let $z = z_2$.

$P(0 < z < z_2) = 0.5 - 0.08 = 0.42$

From the table the value corresponding to 0.42,

$z_2 = 1.41$

$$z = \frac{x - \bar{x}}{\sigma}$$

$$z_1 = \frac{x_1 - \bar{x}}{\sigma} \Rightarrow -0.5 = \frac{45 - \bar{x}}{\sigma} \quad -(1)$$

$$z_2 = \frac{x_2 - \bar{x}}{\sigma} \Rightarrow 1.41 = \frac{64 - \bar{x}}{\sigma} \quad -(2)$$

$$\Rightarrow \quad -0.5\sigma = 45 - \bar{x} \quad -(1)$$
$$1.41\sigma = 64 - \bar{x} \quad -(2)$$

$(2) - (1) \Rightarrow 1.91\sigma = 19$

$\sigma = 9.948$

$\bar{x} = 45 + 0.5\sigma = 49.97$

Mean = $49.97 \approx 50$

Standard Deviation, $\sigma = 9.948 \approx 10$

FITTING A POISSON DISTRIBUTION

- For fitting a P.D. to a given data we've to determine the most appropriate value of the parameter λ.
- This is usually done by making use of the result that λ is the mean of the distribution.
- We take the mean of the given data, $\bar{x}$, as the estimate of λ. Then,

$$f(x) = \begin{cases} \frac{(\bar{x})^x e^{-\bar{x}}}{x!}, & x = 0, 1, 2, \ldots \\ 0, & \text{elsewhere.} \end{cases}$$

- If the total number of observations in the data is N, then the theoretical frequencies corresponding to $x = 0, 1, 2, \ldots$ are $Nf(x)$ (i.e., $Nf(0), Nf(1), \ldots$)

Q: Data was collected over a period of 10 years, showing number of deaths from horse kicks in the 20 army corps. The distribution of the deaths was as follows:

No. of deaths:	0	1	2	3	4	Total
Frequency :	109	65	22	3	1	200

Fit a P.D. to the data & calculate the theoretical frequencies

Ans: Mean of given distribution $= \frac{\Sigma fx}{\Sigma f}$ =

$$= \frac{1}{200}(0 + 65 + 44 + 9 + 4) = \frac{122}{200} = 0.61$$

$\mu = \lambda = 0.61$. (This is the parameter λ of the P.D)

Now,

$$e^{-\lambda} = e^{-0.61} = 1 - 0.61 + \frac{(0.61)^2}{2!} - \frac{(0.61)^3}{3!} + \frac{(0.61)^4}{4!} - \ldots$$

$$= 0.5435$$

Required poisson distribution is $N\frac{e^{-\lambda}\lambda^x}{x!}$; $x = 0, 1, 2, 3, 4$

i.e. $$\frac{Ne^{-\lambda}\lambda^x}{x!} = 200\,\frac{e^{-0.61}(0.61)^x}{x!}$$

$$= 200 \times 0.5435 \times \frac{(0.61)^x}{x!} = 108.7 \times \frac{(0.61)^x}{x!}$$

$\therefore$ Putting $x = 0, 1, 2, 3, 4$ we get,

x	$f(x)$	Theoretical frequency
0	108.7	109
1	66.3	66
2	20.2	20
3	4.1	4
4	0.627	1
		Total = 200

Q: Fit a poisson distribution to the following data which gives the number of yeast cells per square for 400 squares. ($e^{-1.32} = 0.2674$).

No. of cells per square (x)	0	1	2	3	4	5	6	7	8	9	10
No. of squares (f)	103	143	98	42	8	4	2	0	0	0	0

Total 400

Find the expected frequencies.

ans:

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{1}{400}(0 + 143 + 196 + 126 + 32 + 20 + 12)$$

$$\bar{x} = \frac{529}{400} = 1.32$$

$\lambda = 1.32$

The P.D is $N f(x) = \frac{N e^{-\lambda} \lambda^x}{x!}$

$$= 400 \times \frac{e^{-1.32}(1.32)^x}{x!}$$

$$= 106.85 \frac{(1.32)^x}{x!}$$

x	$f(x)$	Theoretical frequencies
0	106.85	107
1	141.05	141
2	93.09	93
3	40.9	41
4	13.59	14
5	3.568	4
6	0.785	1
7	0.148	0
8	0.0244	0
9	0.00358	0

Q: Letters were received in an office on each of 100 days. Assume the following data to form a random sample from a Poisson distribution, find the expected frequencies correct to the nearest unit, taking $e^{-4} = 0.0183$.

No. of letters	0	1	2	3	4	5	6	7	8	9	10
Frequency	1	4	15	22	21	20	8	6	2	0	1

ans:

Mean, $\bar{x} = \frac{\Sigma fx}{\Sigma f}$

$= \frac{1}{100}(0+4+30+66+84+100+48+42+16+0+10)$

$\bar{x} = \frac{400}{100} = 4$

$\therefore \lambda = 4$ $\qquad N = 100$

The P.D is $Nf(x) = \frac{N e^{-\lambda} \lambda^x}{x!}$

$= 100 \times \frac{e^{-4} 4^x}{x!}$

$= 100 \times 0.0183 \frac{4^x}{x!} = \frac{1.83\, 4^x}{x!}$

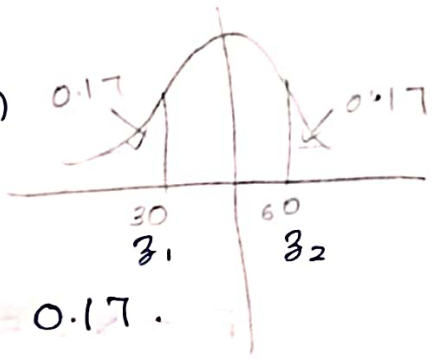
Theoretical freq:

x	$f(x)$	Theoretical frequencies
0	1.83	2
1	7.32	7
2	14.64	15
3	19.52	20
4	19.52	20
5	15.616	16
6	10.41	10
7	5.95	6
8	2.97	3
9	1.32	1
10	0.53	? 1 → 0 or 1

HOMEWORK:

Q:1. In a normal distribution 17% of the items are below 30 and 17% of the items are above 60. Find the mean & SD.

ans: Let $\bar{x}$ and σ be the mean & standard deviation.



17% of items are below 30. means of the ordinate $x=30$ is 0.17.

When $x=30$, let $z=z_1$.

$$P(z_1 < z < 0) = 0.5 - 0.17 = 0.33$$

From the table the corresponding value for 0.33 is $z_1 = -0.96$. (left side).

When $x=60$; $z=z_2$.

$$P(0 < z < z_2) = 0.5 - 0.17 = 0.33$$

From the table the corresponding value for 033,

$$z_2 = 0.96$$

$$z = \frac{x-\bar{x}}{\sigma}$$

$$z_1 = \frac{x_1-\bar{x}}{\sigma} \Rightarrow -0.96\sigma = 30-\bar{x} \quad -(1)$$

$$z_2 = \frac{x_2-\bar{x}}{\sigma} \Rightarrow 0.96\sigma = 60-\bar{x} \quad -(2)$$

(2) - (1) $\Rightarrow$ $1.92\sigma = 30$

$$\sigma = 15.625 \approx 16.$$

$$\bar{x} = 60 - 0.96\sigma = 45$$

Mean = 45

SD = 16

Q: In a test on 2000 electric ~~build~~ bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D of 60 hrs. Estimate the no. of bulbs likely to burn for

(i) more than 2150 hours
(ii) less than 1950 hours and
(iii) more than 1920 hrs and but less than 2160 hrs.

ans: Let x denotes the lifespan of bulb.

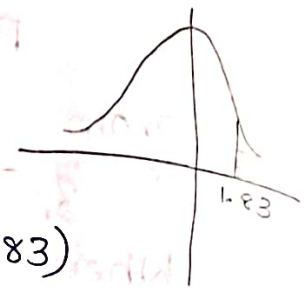
$$z = \frac{x-\bar{x}}{\sigma} = \frac{x-2040}{60}$$

(i) $P(x > 2150)$

When $x = 2150$,

$$z = \frac{2150-2040}{60}$$

$$z = 1.83$$



$$P(x > 2150) = P(z > 1.83)$$

$$= P(0 < z < \infty) - P(0 < z < 1.83)$$

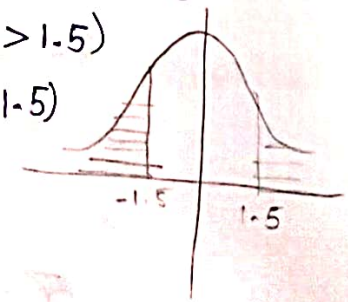
$$= 0.5 - 0.4664$$

$$P(x > 2150) = 0.0336$$

The no. of bulbs likely to bum for more than 2150 hrs is $2000 \times 0.0336 = 67.2 \sim 67$

(ii) $P(X < 1950)$

For $x = 1950$, $z = \frac{1950-2040}{60} = -1.5$



$$P(x < 1950) = P(z < -1.5) = P(z > 1.5)$$

$$= P(0 < z < \infty) - P(0 < z < +1.5)$$

$$= 0.5 - 0.4332$$

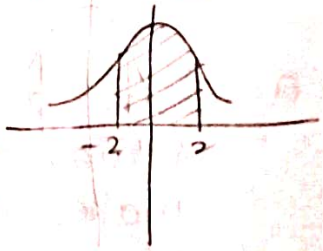
$$= 0.0668$$

The no. of bulbs likely to bum for less than 1950 hrs is $2000 \times 0.0668 = 133.6 \sim 134$

(iii) $P(1920 < X < 2160)$

$x = 1920$, $z = \frac{1920-2040}{60} = -2$

$x = 2160$, $z = \frac{2160-2040}{60} = 2$



$$P(1920 < X < 2160) = P(-2 < z < 2)$$

$$= P(-2 < z < 0) + P(0 < z < 2)$$

$= 2\,P(0 < z < 2)$

~~$= 2 \times 0.5$~~

$= 2 \times 0.4772 = 0.9544$

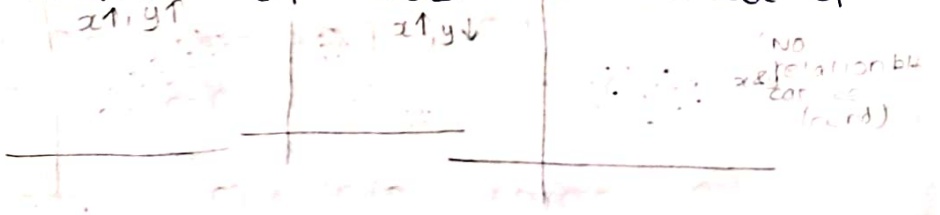
Hence the no. of bulbs that likely to burn more than 1920 hrs but less than 2160 hrs

$= 2000 \times 0.9544 = 1908.8 \sim 1909$

CURVE FITTING AND PRINCIPLE OF LEAST SQUARES

SCATTER DIAGRAM:

- To find a relationship b/t the set of pair observations x and y (say), we plot their corresponding values on the graph taking one of the variables along the x-axis and other along the y-axis, i.e., $(x_1, y_1), (x_2, y_2), \ldots (x_n, y_n)$. The resulting diagram showing a collection of dots is called a scatter diagram.



CURVE FITTING

- Several equations of different types can be obtained to express the given data approximately. But the problem is to find the eqn of the curve of 'best fit' which may be most suitable for predicting the unknown values. The process of finding such an equation of 'best fit' is known as curve fitting. (finding best values of a & b).

- Suppose the points in the scatter diagram appear to cluster around well defined lines or curves, theoretical considerations make us feel that a functional relation exists b/t the variables.
- The form of relation may be suggested by the nature of the curve about which the

points in the scatter diagram cluster or by the theoretical considerations.

Let $y = f(x, a_1, a_2, \ldots, a_m)$ be the suggested relationship where $a_1, a_2, \ldots, a_m$ are the parameters involved.

When $x = x_i$, $y = f(x_i, a_1, a_2, \ldots, a_m)$ if the relation is exact. But in general the corresponding observed value of y, viz, y_i may not be equal to the theoretical value.

It may be subject to some error due to the influence of some extraneous factors. If we take e_i to be the error in y_i,

$$y_i = f(x_i, a_1, a_2, \ldots, a_m) + e_i.$$

Because of the presence of the error term e_i, it is not possible to determine the values of $a_1, a_2, \ldots, a_m$ mathematically.

So some statistical methods are developed to determine the most suitable values of $a_1, a_2, \ldots, a_m$ based on the n pairs of observations (x_i, y_i), $i = 1, 2, \ldots, n$. The process of determining the best values of these parameters statistically is known as curve fitting.

One principle that is must commonly used for curve fitting is the principle of least squares.

We have, $y_i = f(x_i, a_1, a_2, \ldots, a_m) + e_i$; $i = 1, 2, \ldots, n$

$$e_i = [y_i - f(x_i, a_1, a_2, \ldots, a_m)]$$

e_i is called the residual error.

Let $E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - f(x_i, a_1, a_2, \ldots\ldots, a_m)]^2$

E is the sum of the squares of the residual errors.

The principle of least squares states that the best estimates of $a_1, a_2, \ldots\ldots, a_m$ are those values of $a_1, a_2, \ldots\ldots, a_m$ which minimise the sum of squares of residual errors.

Determination of $a_1, a_2, \ldots\ldots, a_m$ by minimizing E can be done by usual methods of differential calculus.

They are obtained by solving the following equations.

$$\frac{\partial E}{\partial a_1} = 0, \quad \frac{\partial E}{\partial a_2} = 0, \quad \ldots\ldots, \quad \frac{\partial E}{\partial a_n} = 0.$$

These equations are called normal equations.

NOTE: This method will be successful only when $n > m$. If $n < m$ the no. of independent normal eqns will be less than m and so there will not be a unique solution.

Fitting of a straight Line $y = ax + b$:

Let $(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$ be the observations. Our problem is to determine the best values of a & b assuming that the given observations are from a population within which x & y satisfy the relation $y = ax + b$.

$$y_i = ax_i + b + e_i, \quad i = 1, 2, \ldots\ldots, n$$

where e_i is the residual error in y_i.

(sum of squares of ...) $E = \sum_{i=1}^{n} [y_i - ax_i - b]^2$ $\left(E = \sum_{i=1}^{n} e_i^2\right)$

By the principle of least squares the best values of a & b are those minimise E, which

which according to calculus are those values of a & b which satisfy the equations,

$$\frac{\partial E}{\partial a}=0 \quad \& \quad \frac{\partial E}{\partial b}=0.$$ ~~(Normal equations)~~ (Normal eqns)

The normal equations are

$$\frac{\partial E}{\partial a}=0$$

$$\Rightarrow \frac{\partial E}{\partial a} \Rightarrow -2 \sum_{i=1}^{n} (y_i - ax_i - b)\, x_i = 0$$

$$\frac{\partial E}{\partial b} \Rightarrow -2 \sum_{i=1}^{n} (y_i - ax_i - b) = 0$$

i.e.

$$\sum_{i=1}^{n} x_i y_i = a \sum_{i=1}^{n} x_i^2 + b \sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} y_i = a \sum_{i=1}^{n} x_i + nb$$

By solving these equations we may get the best estimates of a and b.

By principal of least square, for a minimum of a & b, then $\frac{\partial E}{\partial a}=0$ & $\frac{\partial E}{\partial b}=0$

REMARK

To reduce the normal work involved we may make a transformation of the form

$$u=\frac{x-x_0}{c_1} \quad \& \quad v=\frac{y-y_0}{c_2}$$ where x_0, y_0, c_1 & c_2 are suitable constants and fit a line of the form $v=a'u+b'$. The normal equations will be

$$\sum u_i v_i = a' \sum u_i^2 + b' \sum u_i$$

and $$\sum v_i = a' \sum u_i + nb'$$

$$\sum (y_i - ax_i - b)\, x_i = 0$$

$$\sum (y_i - ax_i - b) x_i = 0$$

$$\sum x_i y_i = \sum a x_i^2 - \sum_{i=1}^{n} b x_i$$

$$\Rightarrow \sum y_i = a \sum x_i + nb$$

06-12-2024 Normal eqns:

$$\Sigma x_i y_i = a\Sigma x_i^2 + b\Sigma x_i \quad \left(\text{from } \frac{\partial E}{\partial a} = 0\right)$$

$$\Sigma y_i = a\Sigma x_i + nb \quad \left(\text{from } \frac{\partial E}{\partial b} = 0\right)$$

Q: If P is the pull required to lift a load W by means of a pully block, find a linear law of the form $P = mW + C$. Compute P when W = 150kg weight.

(Write normal eqns, write table, substitute values in normal eqns, solve eqns, find m & C, substitute m & C in normal eqn)

P	12	15	21	25
W	50	70	100	120

ans: $P = mW + C$

$\therefore$ Normal equations are, $n = 4$

$$\Sigma P = m\Sigma W + nC \quad -①$$

$$\Sigma PW = m\Sigma W^2 + C\Sigma W \quad -②$$

P	W	W^2	PW
12	50	2500	600
15	70	4900	1050
21	100	10000	2100
25	120	14400	3000
73	340	31800	6750

1 25 12 50 1 50 25 3 0 0

① $\Rightarrow$ $73 = m \times 340 + 4 \times C$

$$73 = 340m + 4C$$

$$6750 = 31800m + 340C$$

$$m = 0.188$$

$$C = 2.276$$

$\therefore$ The line of best fit, $P = 0.188W + 2.276$

$(W = 150)$

$P = 0.188 \times 150 + 2.276$

$P = 30.476 \approx 30.5$ kg.wt (unit)

TO FIT A PARABOLA $y = a+bx+cx^2$

- Normal equations are:

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$
$$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

Q: Fit a second degree parabola to the following data.

x	1.0	1.5	2	2.5	3	3.5	4
y	1.1	1.3	1.6	2	2.7	3.4	4.1

Use calculator or transform ($Y = 2x - 5$ & form x, y)

ans:

x	y	x^2	x^3	x^4	xy	x^2y
1.0	1.1	1	1	1	1.1	1.1
1.5	1.3	2.25	3.375	5.0625	1.95	2.925
2	1.6	4	8	16	3.2	6.4
2.5	2	6.25	15.625	39.0625	5	12.5
3	2.7	9	27	81	8.1	24.3
3.5	3.4	12.25	42.875	150.0625	11.9	41.65
4	4.1	16	64	256	16.4	65.6
17.5	16.2	50.75	161.875	548.1875	47.65	154.475

$n = 7$

Normal eqns are:

$$\Sigma y = n a + b\Sigma x + c\Sigma x^2$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$
$$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

Substituting the values,

$$16.2 = 7a + 17.5b + 50.75c$$
$$47.65 = 17.5a + 50.75b + 161.875c$$
$$154.475 = 50.75a + 161.875b + 548.1875c$$

$a = 1.0357 \approx 1.04$

$b = -0.1928 \approx -0.198$

$c = 0.242857 \approx 0.243$

$y = a + bx + cx^2$

~~$y = 1.0357$~~ $y = 1.04 - 0.198x + 0.243x^2$

OR

$y = 2.07 + 0.511X + 0.061X^2$

x	$X = 2x - 5$	y	X^2	X^3	X^4	Xy	X^2y
1.0	-3	1.1	9	-27	81	-3.3	
1.5	-2	1.3	4	-8	16		
2	-1	1.6	1	-1	1		
2.5	0	2	0	0	0		
3	1	2.7	1	1	1		
3.5	2	3.4	4	8	16		
4	3	4.1	9	27	81		

$X = 2x - 5$

$x = \frac{X + 5}{2}$

FITTING OF OTHER CURVES

1. $y = ax^b$ (Power curve)

$$y = ax^b$$

Taking $\log_{10}$ on both sides,

$$\Rightarrow \log_{10} y = \log_{10} a + b \log_{10} x$$

$$\Rightarrow Y = A + bX$$

where $Y = \log y$, $A = \log a$, $X = \log x$

The normal equations are,

$$\Sigma Y = nA + b\Sigma X$$

$$\Sigma XY = A\Sigma X + b\Sigma X^2$$

By solving these we get A and b.
From $A = \log a$, find a.

Q: An experiment gave the following values. Here V and t are connected by the relation $V = at^b$. Find the best possible values of a and b.

V	350	400	500	600
t	61	26	7	2.6

ans:

$$V = at^b$$

Taking $\log_{10}$,

$$\log_{10} V = \log_{10} a + b \log_{10} t$$

$$\Rightarrow V = A + bT$$

where $V = \log_{10} V$, $A = \log_{10} a$, $T = \log_{10} t$

The normal equations are:

$$\Sigma V = nA + b\Sigma T$$

$$\Sigma VT = A\Sigma T + b\Sigma T^2$$

V	t	$T = \log t$	$Y = \log V$	VT	T^2
350	61	1.785	2.544	4.541	3.186
400	26	1.415	2.602	3.6818	2.002
500	7	0.845	2.699	2.280655	0.714
600	2.6	0.415	2.778	1.15287	0.1722
		4.46	10.623	~~11.032~~ 11.656	6.0742

Normal equations, $n = 4$

$$10.623 = 4A + 4.46b \quad -①$$

$$11.656 = 4.46A + 6.0742b \quad -②$$

$$A = 2.8467$$

$$b = -0.171$$

$$A = \log_{10} a$$

$$2.8467 = \log_{10} a$$

$$a = 10^{2.8467} = 702.59$$

$$\therefore \quad V = 702.59\, t^{-0.171}$$

2. $y = ae^{bx}$ (Exponential curve)

$$y = ae^{bx}$$

Taking $\log_{10}$ on both sides,

$$\Rightarrow \log_{10} y = \log_{10} a + bx \log_{10} e$$

$$\Rightarrow Y = A + Bx$$

where $Y = \log_{10} y$, $A = \log_{10} a$, $B = b \log_{10} e$.

The normal equations are,

$$\Sigma Y = nA + B\Sigma x$$

$$\Sigma xY = A\Sigma x + B\Sigma x^2$$

Q: From the table given below, find the best values of a and b for $y = ae^{bx}$ by the method of least squares.

x	0	5	8	12	20
y	3	1.5	1	0.55	0.18

ans:

$$y = ae^{bx}$$

$$\log_{10} y = \log_{10} a + bx \log_{10} e$$

$$\Rightarrow Y = A + Bx \qquad (Y = \log_{10} y,\ A = \log_{10} a,\ B = b \log e)$$

The normal equations are, $n = 5$

$$\Sigma Y = nA + B\Sigma x$$

$$\Sigma xY = A\Sigma x + B\Sigma x^2$$

x	y	$Y = \log_{10} y$	xY	~~$A = \log_{10} a$~~ ~~$B = b \log e$~~	x^2
0	3	0.4771	0		0
5	1.5	0.17609	0.88045		25
8	1	0	0		64
12	0.55	-0.2596	-3.1152		144
20	0.18	-0.7447	-14.894		400
45	6.23	-0.35111	-17.12875		633

$$-0.35111 = 5A + B \times 45$$

$$-17.12875 = 45A + 633B$$

$$A = 0.48118$$

$$B = -0.061266$$

$$A = \log_{10} a = 0.481128$$

$$a = 3.028168$$

$$B = b \log e = -0.061260$$

$$b = -0.1411$$

$$\therefore \quad y = 3.028168\, e^{-0.1411x}$$ (Best curve fitting

$A = \log_{10} a = 0.481128$

$a = 3.028168$

$B = b \log e = -0.061260$

$b = -0.1411$

$\therefore \quad y = 3.028168\, e^{-0.1411x}$ (Best curve fitting

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3. $xy^n = b$ (or $PV^\gamma = k$) - Gas Equation

$xy^a = b$

Taking logarithm,

$\log_{10} x + a \log_{10} y = \log_{10} b$

$\Rightarrow \log_{10} y = \frac{1}{a} \log_{10} b - \frac{1}{a} \log_{10}$

This is of the form,

$Y = A + BX$

where $X = \log_{10} x$, $Y = \log_{10} y$, $A = \frac{1}{a} \log_{10} b$, $B = -\frac{1}{a}$.

Here also the problem reduces to finding the straight line of best fit.

Q: The pressure and volume of a gas are related the equation $PV^\gamma = k$, γ and k being constants. Fit equation to the following set of observations.

P (kg/cm²)	0.5	1.0	1.5	2.0	2.5	3.0
V (litres)	1.62	1.00	0.75	0.62	0.52	0.46

Ans: $PV^\gamma = k$

$\log_{10} P + \gamma \log_{10} V = \log_{10} k$

$\log_{10} V = \frac{1}{\gamma} \log_{10} k - \frac{1}{\gamma} \log_{10} P$

$Y = A + BX$

where $Y = \log_{10} V$, $A = \frac{1}{\gamma} \log_{10} k$, $B = \frac{-1}{\gamma}$, $X = \log_{10} P$

The normal equations are,

$$\Sigma Y = nA + B\Sigma X$$

$$\Sigma XY = A\Sigma X + B\Sigma X^2$$

P	V	$X = \log_{10} P$	$Y = \log_{10} V$	XY	X^2
0.5	1.62	-0.3010	0.2095	-0.0630	+0.0906
1.0	1.00	0.0000	0.0000	0.0000	0.0000
1.5	0.75	0.1761	-0.1249	-0.0220	0.03101
2.0	0.62	0.3010	-0.2076	-0.0625	0.0906
2.5	0.52	0.3979	-0.2840	-0.1130	0.1583
3.0	0.46	0.4771	-0.3372	-0.1609	0.2276
	Total	1.0511	-0.7442	-0.4214	0.59811

$$\therefore \quad 6A + 1.0511B = -0.7442$$

$$1.0511A + 0.59811B = -0.4214$$

Solving these we get, $A = -0.0008747$

$B = -0.7030$

$$\gamma = \frac{-1}{B} = 1.4225$$

$$A = \frac{1}{\gamma}\log_{10} K$$

$$K = \text{Antilog}(A\gamma)$$

$$= \text{Antilog}(-1.2443 \times 10^{-3})$$

$$K = 0.99714$$

Hence the equation of best fit is $PV^{1.4225} = 1.039$